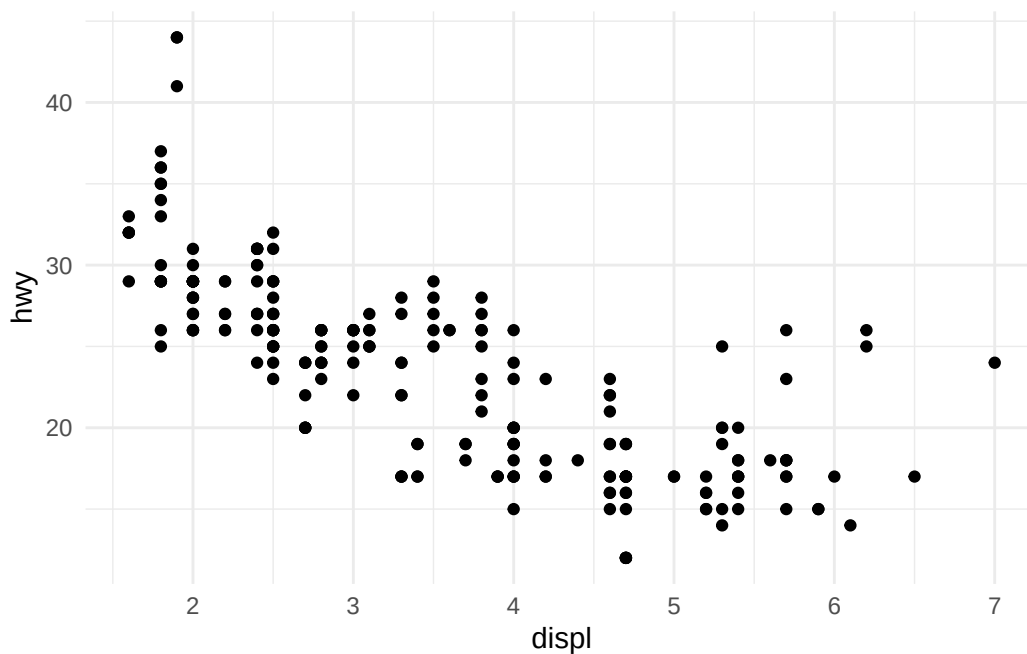


Scatter plot

The scatter plot explores the relationship between two variables.

The basic syntax is given as:

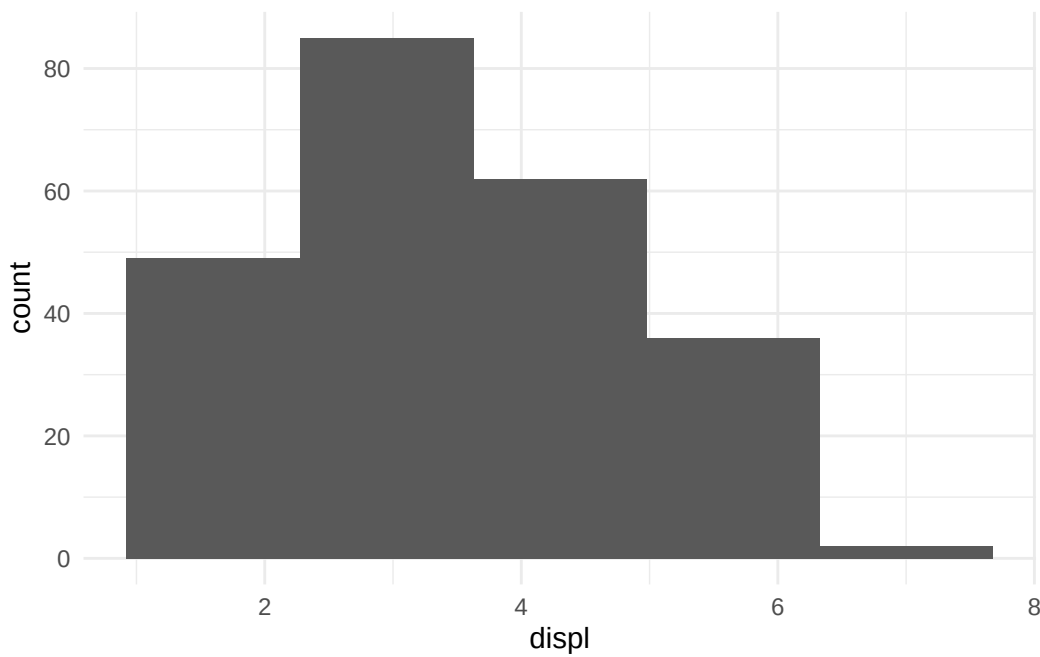
```
ggplot(mpg, aes(x = displ, y = hwy)) +  
  geom_point() + theme_minimal()
```



Histogram

A histogram is an approximate representation of the distribution of numerical data. In a histogram, each bar groups numbers into ranges.

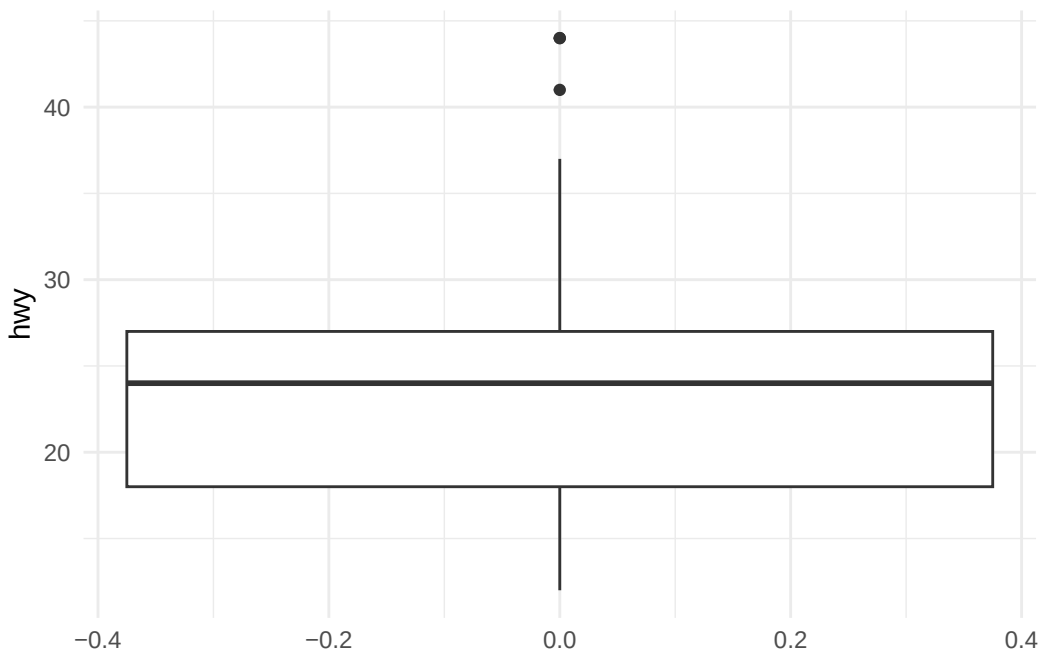
```
ggplot(mpg, aes(x = displ)) +  
  geom_histogram(bins=5) + theme_minimal()
```



Boxplot

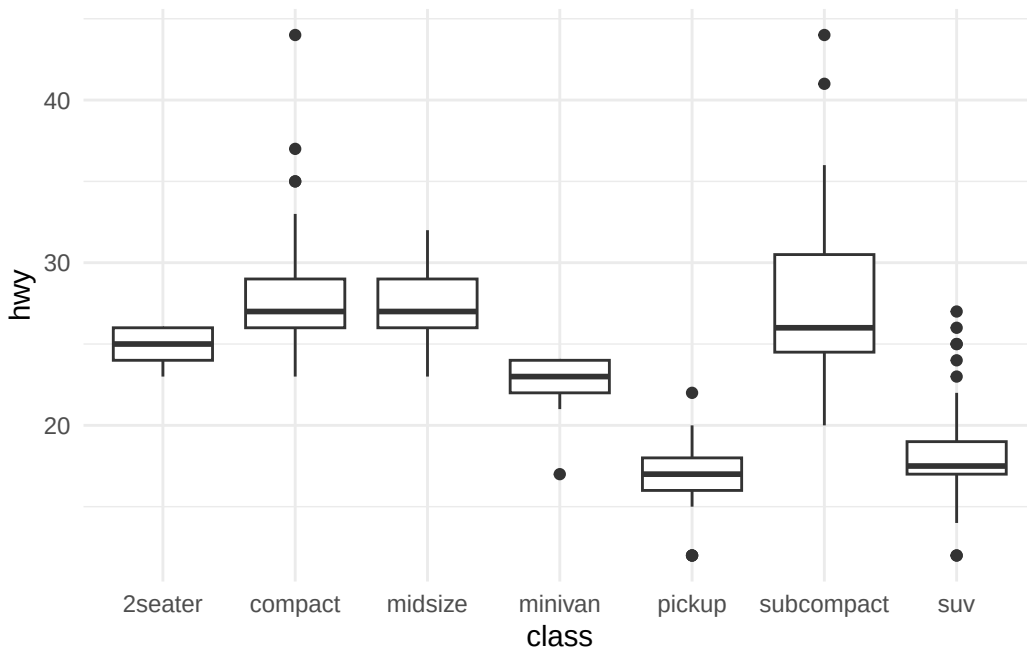
The boxplot compactly displays the distribution of a continuous variable. It visualises five summary statistics (the median, two hinges and two whiskers), and all “outlying” points individually.

```
ggplot(mpg, aes(y=hwy)) +  
  geom_boxplot() + theme_minimal()
```



We can also use box plot to compare a categorical variable and a continuous variable as shown:

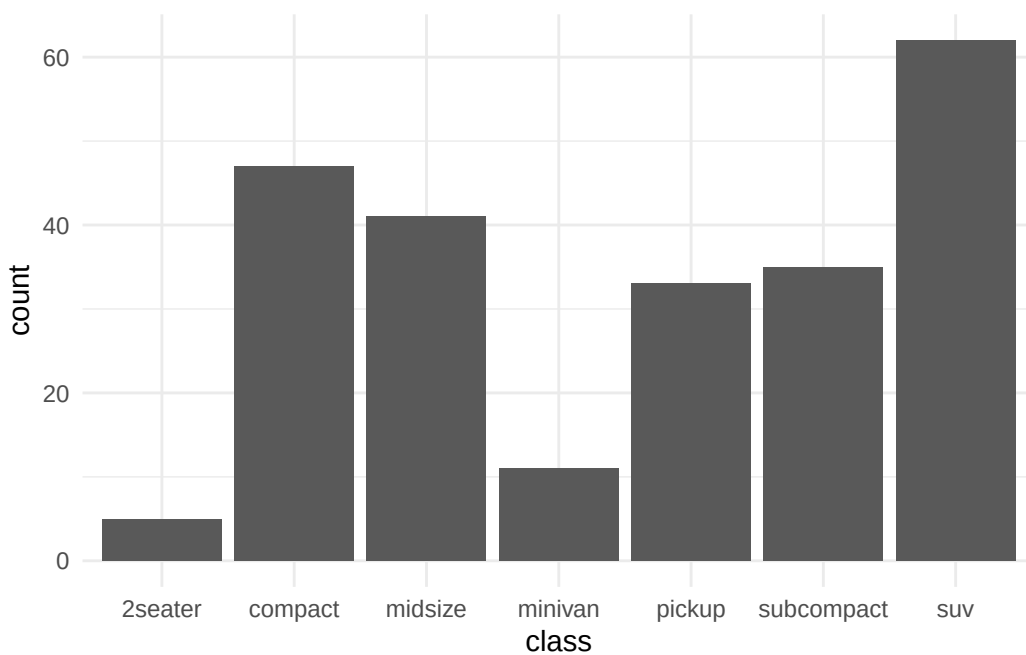
```
ggplot(mpg, aes(class, hwy)) +  
  geom_boxplot() + theme_minimal()
```



Bar Chart

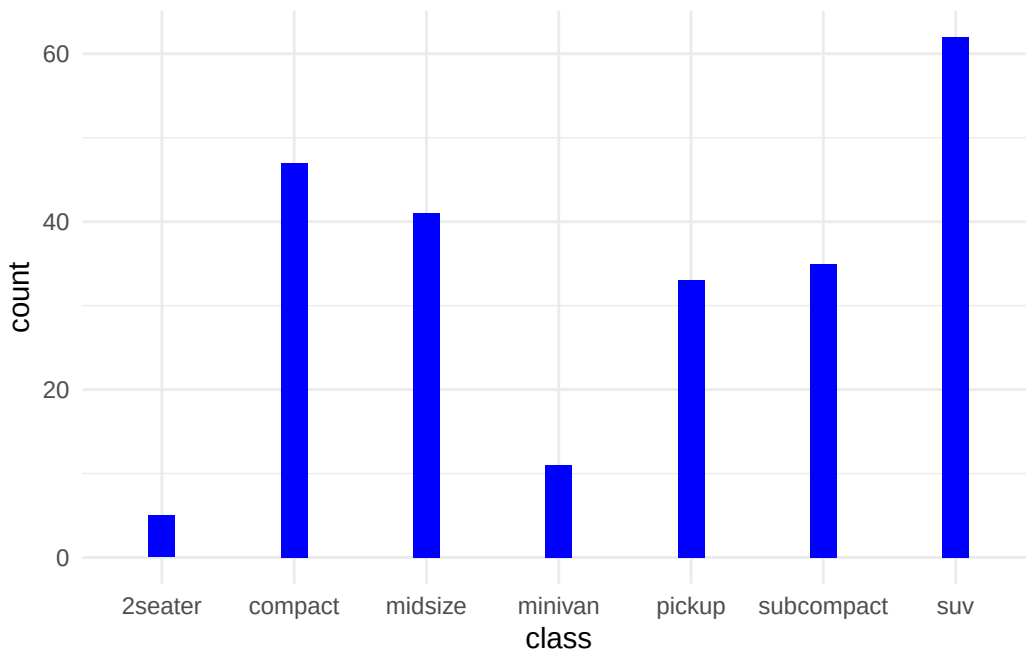
A bar chart is used to visualize categorical variables.

```
ggplot(mpg, aes(x = class)) +  
  geom_bar() + theme_minimal()
```



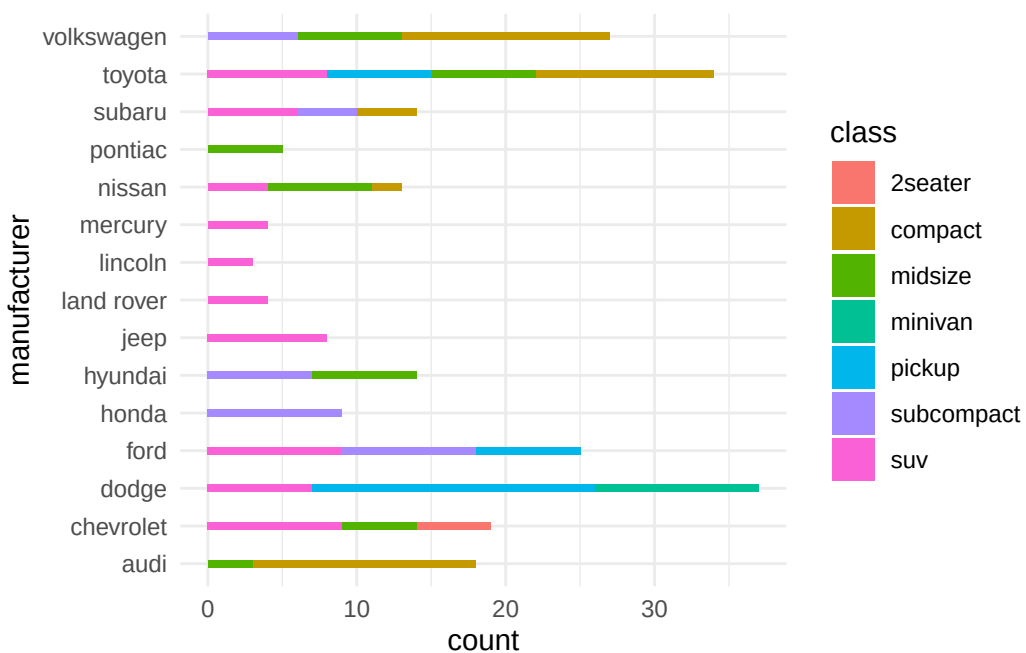
You can control the **color** and the **width** of the bar chart as follows:

```
ggplot(mpg, aes(x = class)) +  
  geom_bar(width = 0.2, fill = "blue") + theme_minimal()
```



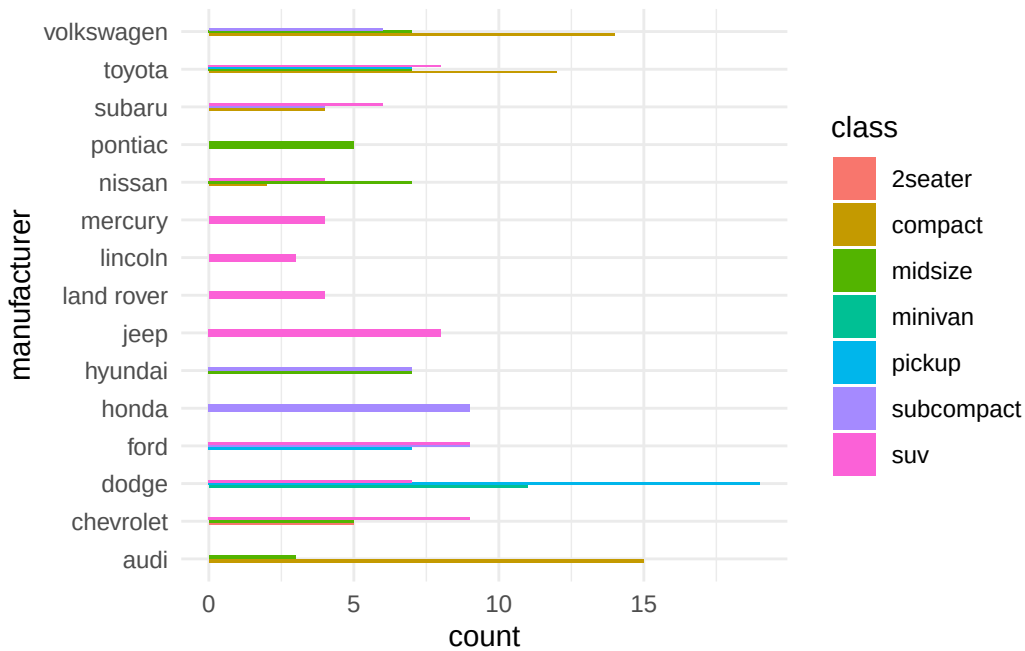
Stacked bar chart We can have a stacked bar plot as follows:

```
ggplot(mpg, aes(x = manufacturer, fill = class)) +  
  geom_bar(width = 0.2) +  
  coord_flip() +  
  theme_minimal()
```



Clustered bar chart We can have a stacked bar plot as follows:

```
ggplot(mpg)+
  geom_bar(aes(x = manufacturer, fill = class),width = 0.2, position = "dodge") + coord_flip()
```

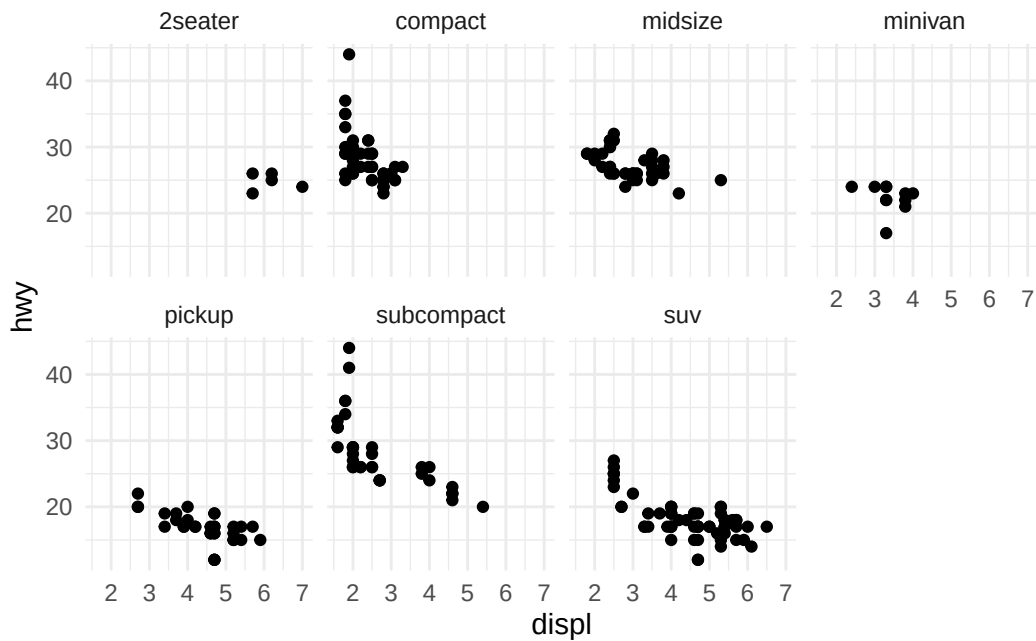


Facets

One way to add additional variables is with aesthetics. Another way, particularly useful for categorical variables, is to split your plot into facets, subplots that each display one subset of the data.

To facet your plot by a single variable, use `facet_wrap()`. The first argument of `facet_wrap()` should be a formula, which you create with `~`

```
ggplot(data = mpg) +
  geom_point(mapping = aes(x = displ, y = hwy)) +
  facet_wrap(~ class, nrow = 2) +
  theme_minimal()
```



Labelling Charts

We can modify the axis labels and title using the `labs()` option in the plot as follows:

```
ggplot(mpg, aes(x = class)) +  
  geom_bar(width = 0.2, fill = "blue") +  
  theme_minimal() +  
  labs(x = "Type of Car",  
       title = "Bar graph of type of car")
```

